

Lec 62 Prompt Engineering Basics

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Transcript

[00:00] Hello everyone, welcome back to the lecture series on fundamentals of generative AI and large language models. In our previous lectures, we have discussed about GPT bird and GPT and in today's session we will be discussing about prompt engineering which is a completely new topic and how to design effective prompts for large language models. So that is the discussion uh we will be having on in today's session. So we will look into the contents uh the contents goes like the concept of prompt engineering why prompts matter techniques of prompting under that one we will study zeroshot prompting oneshot prompting few short prompting chain of thoughts structured output prompting and role prompting. Okay. And then finally

[00:50] we will go into applications of prompt engineering. Okay. So we will start with a concept of prompt engineering what exactly it is. Okay. So suppose you are an user okay of a large language model and you want to use that LLM for giving some answer to some of your queries. Okay. So for that one you need to tell to the model what exactly is you want the model to tell. Okay. Suppose you want the weather of say Bangalore city you know some 10 years back. So you have to tell the model that this is my question. Okay. And that is how you pose the question to the model. That exactly is called the prompt. Okay.

[01:42] Prompt or the input. So prompt is the input given to an AI model that guides its response. Okay. And then this prompt goes to the language model large language model and the large language model gives the AI generated response. So that is what what exactly the prompt means. Now what exactly is prompt engineering all about? Prompt engineering is a topic or a domain of study where we discuss about or we learn how to design effective prompts for giving it in LLMs.

[02:21] When I tell effective prompts, what do I mean? I mean that the question should be put in such an effective way that the answers that the LLM gives us is what we want what we expect the LLM to give. Okay. So it is what exactly is the question that answer I get what what information the user wants that information only the LLM gives rather than giving much broader or much you know um lot of information or less information. Okay. So this prompt engineering is a complete domain of study where we see that better prompts lead to better responses. So the different things that comes in under prompt engineering is basically how to write an effective prompt. What are the different techniques by which we can

[03:12] tell the LLM that this is this kind of answer I want. Okay. So that is what is about prompt engineering. And if you say a prompt may include what are the things that a prompt includes? So it includes some instructions, some questions, some context, some constraints and some examples. Okay. So the more you give in the prompt, it becomes much more clearer to the LLM what exactly you want. Okay. So suppose you have an example like summarize this article in five bullet points. So this is basically you are giving some instruction. Okay.

[03:53] Instruction means what is the instruction? You are telling it to do some summarization. [snorts] You are giving five bullet points. That is also an instruction given. Summarize this article. Now before this one, you must have given some context. Some article you have put as an input to the LLM. So that becomes your context. Okay. And from after this one you can write based on this you know summary tell me what was the uh you know weather condition in Bangalore in last 5 months. Okay. So that is a question you can give some constraints in that in your input as well like do avoid including rainy seasons. Okay. So I want information only about summer, winter and other other seasons not the rainy season. So

[04:48] these kind of constraints you can give. Okay. And also some examples you can put. So that's that all these things can come inside your prompt and more and more and more you know clarity of what exactly you want the LLM to give you answer about the the LLM gives better response. Okay. So the clarity of the prompt is very important. Now once again why prompts matter. So in prompt when we give a prompt to a uh large language model the prompt tells whether what is the task that is giving. So defining the task. So the prompt is used for defining the task like summarization, translation, explanation, classification or question answering, generation of code anything. Okay. Now that is task definition. So that is a part of your

[05:42] prompting provide context. So you give some background information that is this background information is some that is what we call it as context which helps the model to understand the situation and produce you know more meaningful results relevant responses. Okay. So the context matters. So you have to give some context context. Okay. Under this scenario you have to tell. Okay. So that is something which uh gives the LLM a much better clarity on how it will put its response. Okay. Specify the audience. This is very important. If you specify the audience, the LLM understands how much of information you it needs to put in that answer to the prompt. Suppose if you are putting we will see these examples in my next

[06:38] slides also. Suppose you are trying to explain a concept of AI to a layman. The answer of that prompt will be different from if you tell that I want to explain this concept to PhD scholars or research scholars the depth of the answer that is coming out of the LLM will be different. because these are two completely separate kind of audience. Some will have very limited technical knowledge and the other one will have quite a bit of technical knowledge. So the answer that is coming out of the LLM will be very specific if you define the audience. Okay. So that is very important.

[07:26] Controls the output format. So this is also uh a very significant uh part it plays. So suppose the output you want in bullet points you want the output in the form of table. If you want the output you know in the form of code or essay or anything it will give you in that format. Suppose you need to write three bullet points and the answer that is coming out of the LLM is a big paragraph that is not going to help right. So in that case we can tell we can prompt the LLM to give me an output in bullet points three bullet points five bullet points like that. Okay.

[08:07] Then the other thing is reduce the ambiguity. When we give give a prompt, if we specify the task, give some context, specify the audience, tell it in which format it should give me as an essay or a bullet point or a summary, anything or a quote a code snippet in the form of a code snippet. Okay. All these things when you give it reduces the ambiguity. Together with all these things if we give some more specific instructions it the it helps the model to understand the context better understand the question better and misunderstanding and misinterpretations can be avoided a lot.

[08:53] Okay. So that is reducing ambiguity. So better prompt reduces ambiguity and it helps in uh making the LLM give much better responses. Okay. So that's that's all why prompt matters. Then we will look into characteristics of good prompts. So what do you what do you expect a good prompt to look like? Okay. So it should be clear very clear crisp and clear what exactly you want. It should be context rich. You give some you know pre uh required information to the you know um model.

[09:36] Suppose you want something in the prehistoric era. Some information from the prehistoric era. You have to give some context. How was the situation? How was the political situation? How was the social you know situation that time? Okay.

And then you ask it to give you some information. Okay, some question unambiguous very important what exactly you want. You want you know um a very specific answer you ask that one.

[10:04] Don't make it you know a broad one or a very you know diffused question kind of thing very goal oriented and very specific. Okay. So these are the things which are the characteristics of a good prompt. If you want your LLM to give the answer for um say what is the situation in Bangalore or what is the political situation in Bangalore a week before. Okay. So in that case you are specifying the time a week before you are specifying which city you want in India whose political situation you want instead of giving give me a political situation in India instead of that one you are very specific about the location you are specific about the duration you are specific about what exactly you want in

[11:04] politics there may be so many things or in sports there may be so many things so you can tell what is the update of cricket in my in the last 7 days. So you are specifying okay what is the sports update in last 7 days that will give you some you know vague answer if you specify like give me the sports update in the last 7 days played by cricketers who are in Karnataka. Okay so something in that direction you make it very specific. Okay that those are the uh characteristics of good prompts.

[11:40] What are the things which we should avoid? Vague instructions like like you know you yourself don't know what exactly you want the answer to be missing context. If you don't give a pre required context then what will happen is LLM can start hallucinating and it can make out its own context and then give you some answer which will not be proper. Also don't give multiple unrelated questions. What is the weather of India? How is the political situation in India? What is means what is the sociological or what is the demog what is the weather condition in this this part of India? So many unrelated questions if you put then the model gets confused and it will give you some response. Of course it will

[12:31] give you some response but that may not be the you know best one. Okay. So avoid vague instructions missing context. You have to give some context. Suppose you want some uh responses about diabetes. I'm just telling. So you give some context some medical you know uh document to the uh LLM and which is related to diabetes. Suppose you have some scientific documents on diabetes, its symptoms, its treatments and other things. Put that one in the LLM and then that is the context and then based on that context you ask the questions. What are the symptoms? Okay. What are the symptoms of diabetes? How do I control these? What are the treatments? What are the you know um um what are the Yeah. What are the

[13:24] symptoms that the student the the patients face? what are the most challenging part in this entire journey of diabetes. So once you give the entire context from a very relevant scientific literature or a medical report then you ask the model to give

you answers then the answers will be much more precise much more accurate. Okay. So that is something we must give some meaningful or very um trustworthy context to the LLM. Okay.

[14:00] Then we will just have a look into what is a vague prompt and what is a very specific prompt. I was telling vague instructions or vague prompts. So suppose the prompt is explain machine learning. This is the prompt you are giving to the LLM. The response can be machine learning is a branch of AI that enables computers to learn data without being explicitly programmed. It includes supervised learning, unsupervised learning, reinforcement learning and many algorithms such as decision trees, neural networks and support vector machine. So this is one of the possible answers responses. Now what is the problem with this one? This do we have a problem with this kind of response?

[14:43] The problem that may occur here is there are lot of scientific terms okay technical terms basically supervised unsupervised algorithms decision trees neural net so your your response from the LLM is completely or heavily uh you know uh it is heavy on technical words heavily loaded with technical words now suppose because that response is read by a newcomer in that domain. They will not get much of information out of this response because half of the things are not known to them. Okay. So if you give a prompt like this explain machine learning these kind of answers may come and the response is very broad and it may be too detailed. I may not need to know so many details supervise unsupervised reinforcement

[15:42] that might not be my um that not be the information that I'm asking you know or looking for okay I want some you know high level understanding of machine learning so unless and until I give a very specific prompt to the LLM this kind of vague not vague very broad too too much details or too less details our responses may come out. Okay. And that will not help the intended audience. Okay. Now what is the example of a specific prompt? So now I change that prompt explain machine learning to explain machine learning to first year engineering students in 150 words. Use simple language and include one real world example.

[16:32] If that is the prompt, what answer do I expect the LM to give? Machine learning is a this is one of the possible answers. Machine learning is a type of artificial intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed for every task. For example, this is an example, you know, real world example. For example, an email application can learn to identify spam messages by analyzing thousands of previous emails.

[17:07] As it processes more data, it becomes better at distinguishing spam from legitimate emails. Okay, so this is the answer. What are the key points? Key take-aways. Learns pattern from data. Improves performance with experience. Used in applications such as spam filtering, recommendation system, and image recognition.

So here you can see if a student first year engineering student reads this response from the LLM then it will be that student will be able to understand couple of things. One is what exactly is machine learning with an what exactly is machine learning and an example of that one. Once he or she is trying he or she is able to understand the example the things become much more clearer and then the final key

[17:59] points. So this becomes much more specific the answer becomes much more target oriented target audience oriented. Okay fine. So more and more we go to specific prompts more and more the response from the AI model becomes useful. Okay. Why it is better a response matches the audience, the length and the format requested. Okay. So all these things are taken care in a specific prompt. Then we will see what are the different techniques of prompting. Are there any different techniques? Yes, there are a couple of different techniques. You know B in basic prompting techniques. One is zeroshot prompting, one short prompting, few short prompting, chain of thoughts, structured output prompting, role prompting. Okay. So these are the

[18:50] various techniques of prompting. Now we will look into one of them each. What is zeroshot prompting? Zeroshot prompting from the name itself there is a term zero. Okay. So the LLM performs a task without being provided any examples. So zero examples are shown to the LLM for performing a specific task. So this is called this type of prompting is called as zeroshot prompting. So in this case I don't give any example to the LLM. I just rely solely on the instructions in the prompt and the pretend knowledge that is already there in the LLM. Okay.

[19:33] So that those two things. So two things that we um uh see one is instruction in the prompt and the pre-trained knowledge of the LLM. So based on these two only we expect the LLM to give the response. We don't give any specific example for that kind of task. So no training examples are included in the prompt. The prompt contains only task specific dis uh description and the model infers the required output based on the learn learned knowledge. So this is an example. Classify the sentiment of the following sentence as positive, negative or neutral. The movie had stunning visuals but the story line was disappointing. So the LLM's response will be negative. And here the LLM have given me the sentiment analysis of this

[20:28] statement as negative based on its predent knowledge. So whatever knowledge it has it has used that one and come up with this response. So this is called as zeroshot prompting. Okay, one short prompting from the name itself we can make out once one example is shown to the model to get the output to get the response from the model. Before you give the task to the model, you show one example. Okay. And why you want to show that example to the model? Because you want the model to get an understanding of what is the task that is on hand.

[21:12] Okay. And what is the expected answer? So we give one input output example to the LLM and then expect it to give for other uh input output. Okay. Now example.

So if you give a prompt like the customer support was excellent, the sentiment is positive. That is one example. So this is your example. So you are giving this example to the model and then the this one basically is your task to be performed. The laptop battery drains very quickly.

[21:57] So what is the sentiment? So this is the question you are asking to the model. Before you ask the question, you have given an example only one example that is why it is called one shot. Only one example you will give and then it will you will ask the LLM to give the response for another such you know prompt another such sentence. So this type of prompting is called as oneshot prompting. So here based on this example the laptop battery drains very quickly the result will be negative. Okay. So that is what is about one short prompting. Then we come to few short prompting. Once again from the name itself there is few that means few examples are shown to the model before the actual task is asked.

[22:50] And once again it these examples help to understand the desired format style the reasoning pattern which leads to more accurate response. Okay. So this is what it is and so here you give two to five examples and let the model learn like here. So here you have given two examples. I absolutely loved the food. Sentiment is positive. I served the service was very slow. Sentiment is negative. So these are two examples given.

[23:22] given as a part of your prompt. Okay. And then this is the task that you are asking. The phone was has had a good camera but poor battery life. Okay. So what will be the sentiment? So this is this kind of prompting technique is called as few short prompts. Okay. So we have understood zero zero short oneshot and few short-shot prompt. Then we will go into applications of zero short oneshot and few short prompting. Where are these applications? So zeroshot applications for z the applications for zeroshot prompting are take summarization translation sentiment analysis question answering grammar corrections. Okay. So these are the typical examples of zeroshot prompting. One short prompting it is teaching a specific response

[24:17] format maintaining a writing style. Suppose you want to write an answer like Shakespeare. So you give some you know dialogues from some Shakespearean drama and then asked to write such you know similar dialogues in that similar uh style Shakespearean style. Okay. So this is one short prompting. You give example and then ask uh explaining customer labeling, generating structured outputs. All these things come under uh or or uh in these cases oneshot prompting is used and few short prompting is used for customer text classification, information extraction, domain specific task, reasoning task and intent classification. So customer text classification. So if this kind of uh you know text comes from the customer it

[25:15] means they are happy customers. This kind of text coming okay the the the product was very good. The last u product that I received was very good. So this type

of text if it is coming from any customer that is a happy customer. Then if you have something like the product was extremely bad that is a you know angry or dissatisfied customer. So this kind of examples if you give to the model before you ask it to classify some text that is coming from customers then it will be easier for the model to do so. Okay. So these are the few exa application areas of few short uh prompting. Then we come to the another um prompting technique which is called as chain of thoughts.

[26:02] What is this chain of thoughts? Basically chain of thoughts is a technique which is used for solving some numericals inside your by the LLM. Okay. So we'll just give see uh the example of a prompt and the LM response. So if you have some calculations to be done then we go for this chain of thoughts prompting because calculations require stepbystep thinking. Okay. And that is where your chain of thoughts comes into picture. Okay. So there is a sequence of steps through which if you go then you get the output. Okay. So we'll just take an example. See a shop sells a notebook for \$5 and a pen for \$2. The customer buys three notebooks and four pens. Calculate the total cost. Think step by step. So

[26:56] this is what is there is a shop and there is a you know shop uh you know there is a customer here. There is a shopkeeper here. There are notebooks and pens here. So, notebooks are \$5 and pens are \$2. Now, customer buys three notebooks and four pens. And then I need to my LLM needs to give me the total cost. Now, this requires some calculations. So, what will be the LLM's response in this case? Step one, cost of three notebooks is 3 into \$5, which is \$15. Cost of 4 P is 4 into \$2 which is \$8. Total cost is \$15 plus \$8 which is 23. So the final answer is 23. So whenever there is a stepbystep reasoning stepbystep calculations or reasoning you know basically like okay if one book is notebook is \$5 then

[27:55] three notebooks will be \$15. So these are all reasoning steps that are coming. Okay. So from here from these steps the LLM can give the response properly. Okay. So this kind of prompting is called as you know chain of thoughts. Okay. Fine. Then [clears throat] role prompting. Role prompting is what is role prompting? You are telling the model to mimic the role of a certain profession. That can be a doctor, that can be a nurse, that can be a teacher, that can be an athlete, that can be a nutritionist, that can be a musician, anything, an artist, anything. Okay? So, you define a role for a particular task and tell that like a teacher, you explain this one. Like a musician you tell me what are the steps I should

[28:54] follow so that I can compose music well. Like a athlete you give me what are the routines that I should follow so that I can build strength. Okay. So like this if you give a specific role you define a specific role to your LLM then it becomes much easier for the LLM to give you better response. Like there is an example. The prompt is you are an experienced physics professor teaching firstear engineering students. Explain Newton's second law in simple language with real life examples.

[29:32] So this is the prompt. So you are actually telling the role experience physics professor here you are defining the role. First year engineering students here you are defining the audience explain Newton's second law of Newton's second law this is defining the task with real world examples or something okay so this is basically role prompting okay and what will be the LLM's response. Newton's second law states that the force acting on an object equals its mass multiplied by the acceleration $F = ma$. For example, pushing an empty shopping cart is easier than pushing a fully loaded cart because the heavier cart has greater mass. So, it requires more force to achieve the same acceleration. So, we

[30:36] have an example here. This is the real world example. And like a experienced physics teacher this is the this is the response. So very crisp very compact very to the point answer. Okay. So this is about role prompting. Now this this can be for some something as a experienced dietician give me a diet to be followed for next one month so I can increase my uh muscle strength or I can reduce my fat content in the body or I can reduce my weight. Okay. So as an experienced nutritionist so that is the role that you are specifying the LLM to give response for. Okay. So that these are the things which comes under role prompting.

[31:27] Then there is something called as the structured output prompting. What is this one? That means you are telling the model that this kind of structure I want the response. In this structure I want the response. It can be bullet points. It can be JSON file. It can be uh as a summary. It can be as uh you know something else as an essay. Okay. So you tell what exactly you want. What is the format you want your answer to be in. So extract the information from a paragraph below and return the output as a JSON object. Okay. So John so what is the text you are giving? John Smith is a data scientist at ABC technologies.

[32:06] Okay. So this is the text that you are giving and you want the JSON file to be created for this one or output to be in the form of JSON. Okay. So name is so this is the specific format output. Okay. And you can like this you can tell you give me the output in the form of bullet points. You give me the output in the form of a small uh summary of say uh 50 words anything. So whatever you want okay or give me I want the code snippet for this one. That means you require some code to be put as coming out as the from the LLM. Okay. So this is about structured output prompting. So you are telling which which structure or which format you want the prompt to be in. Now applications of prompt engineering. What

[32:58] are the applications? One is context generation, data analysis, content generation. So you give a prompt then from there you can uh um generate images, you can generate uh videos, you can generate anything data analysis. Yes, you give proper prompts for data analysis and get what exactly what analysis you want. Last few weeks analysis, last few months analysis, whether you want the bar graph, the pie chart, what what what you want, whatever you want, you have to give that one.

[33:28] Programming assistance, education, healthcare, customer support, you know, everywhere, research, document summarization, translation, in every domain. Now today, we require this prompt engineering. We require proper prompts to be given to the LLM so that we get proper responses. Okay, better responses. So with this one I come to the end of my today's session. So before concluding let me summarize what we learned. We have learned about prompt prompt engineering today. In today's session that is the how the effect how are the what are the effective ways by which we can tell the model to give me answers which better suit my expectation. Okay. So in that category we have understood what are the

[34:15] characteristics of good prompt. What are things to be avoided? Vagueness, you know, context missing or you know multiple un unrelated questions. These things should not be there in a good prompt. Good prompt should be clear, crisp, have the context, very specific goal oriented questions. So these are the things of a good prompt. Then we have seen how are the different prompting techniques uh like one short, few short and uh zeros short prompting where we give uh either zero examples, one or few examples to the LLM to understand what exactly is the style or what exactly is the task we want them to do and then based on that uh uh one example or few examples that we are giving we ask the LLM to perform a

[35:02] specific task in that similar uh pattern. Okay. Then we have seen there is something called as the chain of thoughts which are actually required for some calculation oriented prompts. Okay. So you give them step by step and then tell them to you know think step by step and get the answer response. Then we have seen structured prompts where we tell that the structured output prompt that where we tell the model to give us output in a particular format. So all these things are different prompting techniques very basic different prompting techniques. Then we have seen what are the applications of these kind of prompts where all we can apply and uh where all we can apply in general pro prompt engineering. It is like every domain

[35:50] nowadays or today every domain requires better uh you know prompt engineering because better the prompt better is the response. You want very very scientific details about any uh research paper or something. You give the context and you get a lot of information. You want some very uh high level or very system level knowledge of something. You ask or give you you design your prompt in such a way that it gives you only you know high level knowledge not going into the details. So prompt engineering is a extremely extremely uh important topic in nowadays to understand uh how to get the best response from your LLM. Okay.

[36:32] So with this one I come to the end of today's session and this is my last session in the uh u lecture series on fundamentals of generative VI and large language models. So uh with this session I conclude my entire lecture series. After this there will be few more sessions um on uh you know rags and other things. So please keep learning and enjoy this course. Happy learning and thank you all.